Machine Learning

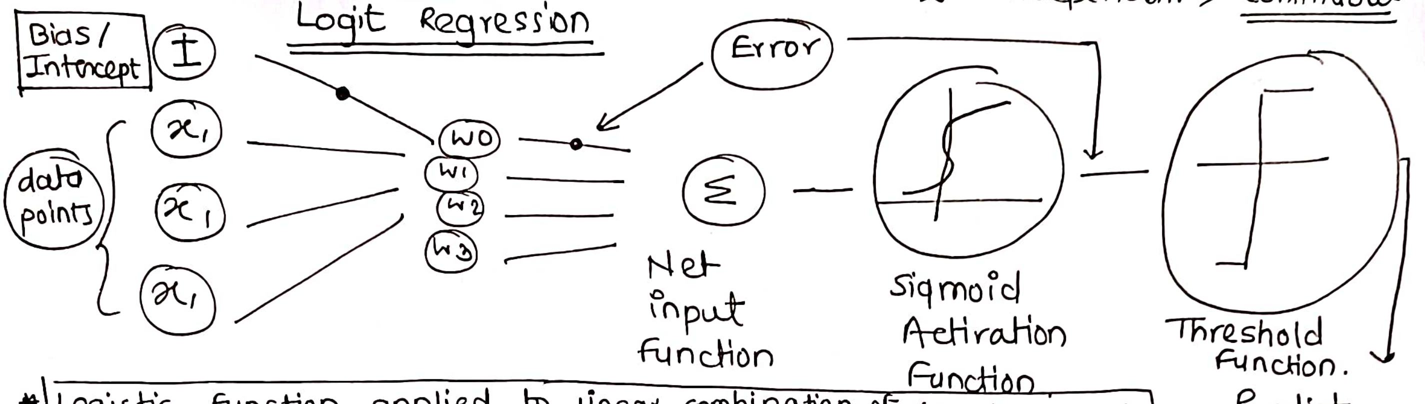
Both → Classification, Regression

Logistic Regression :-

$Y = mx + C$

y = dependent ⇒ Binary (0,1)
x = independent ⇒ Continuous

Logit Regression

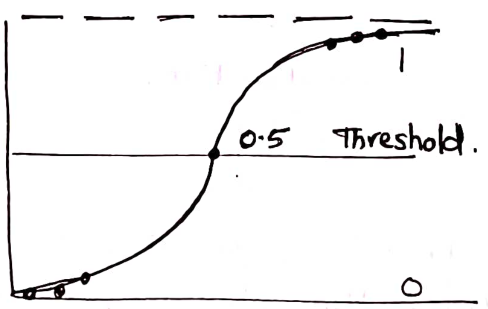


Predict class label

* Logistic function applied to linear combination of input features to map the output to probability 0 to 1.

W = weight matrix or vector

The goal of logistic regression to predict the likelihood of $y=1$ rather than $y=0$.



sigmoid activation function compared with threshold to decide predicted class label.

$\boxed{R^2}$ = fitness of overall model

$\boxed{\text{Chi square}}$ = How regression fit data

$$\boxed{\text{Optimization} = \text{Loss Function} + \text{Regularization}}$$

1. More – Accept feature.
2. Less – Give chance to other function features.

Adjust loss function thus minimize loss function.

* **Activation function** :- $\sigma(z) = \dfrac{1}{1 + e^{-z}}$ ⇒ Mapping ~~the probabilit~~ the predicted value to probability

Sigmoid / Logistic

$$\text{Minimize} = \frac{1}{m} \text{Log of } (1 + e^{-z}) + \underbrace{\lambda}_{\text{Hyper parameter}} \underbrace{\|w\|_2^2}_{\text{Regularization}}$$

where $\boxed{z = ywx}$ y = output w = weight x = Feature

Loss ⇒ It is miniest at value ∞.

* A logistic regression model predicts a a dependent variable by analyzing the relation between one or more existing independent variable.

$$\boxed{\text{minimize}} = \frac{1}{m}\underbrace{\sum_{i=1}^{m} \log(1+\exp(-bAx_i))}_{\text{Loss Function}} + \underbrace{\lambda \|x\|}_{\text{Regularization factor}}$$

Loss Function (It is minimum at ∞)

Regularization. factor (Resistance)

$\boxed{\lambda}$:- Hyper parameter. It have value between 0 to 1.

Objective :- To minimize the loss function

✓ overfit :- When model trained on outlier. $\boxed{\text{More overfit} \Rightarrow \text{More variance}}$

✓ underfit :- Mis-classification. pathetic model training.

$\boxed{\text{More underfit} \Rightarrow \text{More biased}}$

∴ ✓ overfit ⟶ variance

✓ underfit ⟶ bias.

$\boxed{y = mx + c}$ $\boxed{mx = \text{coefficient},}$ which feature is more important. $\boxed{c = \text{Intercept}}$.

$\Sigma = m_1x_1 + m_2x_2 \dots m_nx_n$

* Plotting Decision Boundary. - line of classification.

- Precision - Ratio of True positive to Total predicted positive
- Recall - Ratio of True - // - pred to Total actual positive
- F_1 Score - Harmonic mean of Precision & Recall.
- Support - How many time or frequently data appeared in model.

L_2 Norm of the vector :- (Ridge)

Adds square magnitude ✓

Q) **Difference between logistic Regression & Linear Regression**.

* Linear Regression - It provides continuous output - Infinite output
* Logistic Regression - It provides constant output. - few output

* IF No. of observations is lesser than No. of features, regression is not used because of overfitting

* Logistic Regression required average or No multicollinearity between independent variables.

* IF x & y have strong positive relation linear relation probability of y = 1 increase

* A Perfect relationship represents a perfectly curved S rather than a straight line.

$\boxed{y=1 \rightarrow \boxed{P} \qquad y=0 \rightarrow \boxed{1-P}}$

$\ln\left(\frac{P}{1-P}\right) = a + bx$

$\boxed{\ln\left(\frac{P}{1-P}\right)} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots \varepsilon$

ln = natural logarithm and $a+bx$ = regression line.

Expected probability $y=1$ for given value of x $= \boxed{\dfrac{\exp(a+bx)}{1+\exp(a+bx)}} = \boxed{\dfrac{e^{a+bx}}{1+e^{a+bx}}}$

1) **Sigmoid function** (Activation)

2) **Log - odds (Logit function)**

The log odds (logit) of the probability of positive class (1) as a linear function of input variable.

* This linear relationship allows logistic regression to model the probability of binary outcomes.

3) **Predicted Probability**

The probability of the positive class (1) given by

$\boxed{P = \dfrac{1}{1+e^{-z}}}$

If $P > 0.5 \longrightarrow$ class 1 (positive)

If $P < 0.5 \longrightarrow$ class 0 (Negative)

4) **Loss Function** :- Log - Loss (Cross - Entropy Loss)

This loss functions measures how well the model's predicted probabilities match the true binary labels.

$$\boxed{\text{Log - Loss} = \frac{1}{N}\sum_{i=1}^{N}\left[y_i \log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)\right]} + \lambda \sum_{j=1}^{r} \beta_j^2$$

✓ y_i = actual label (0 to 1) for ith sample.

✓ $\hat{y}_i$ = predicted ~~label~~ probability for ith sample.

* for $y=1$:- The loss is $-\log(\hat{y})$, which penalize the model if the predicted probability $\hat{y}$ is low

* for $y=0$:- The loss is $-\log(1-\hat{y})$, which penalize the model if the predicted probability for class 1 is high.

lower the log - loss - better the model.

5) **Gradient Descent** :- To minimize the log-loss function.